

Fundamentals of Computer Programming (I)

Homework - Lecture 1

1. What is the essential difference between programming languages and algorithm description languages?

4. Given the piecewise function:

$$y = \begin{cases} x^3 + x + 1 & x \geq 0 \\ x^3 - x - 1 & x < 0 \end{cases}.$$

Describe the process of calculating and outputting the function value after inputting an x value using natural language, algorithm description language, and structured flowchart respectively.

5. Given the polynomial:

$$s = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \cdots + \frac{(-1)^n}{2n+1}x^{2n+1}.$$

Describe the process of calculating and outputting the polynomial value after inputting x using natural language, algorithm description language, and structured flowchart, until the absolute value of the last term is less than 0.000001.

6. Following Example 1.1, write a C program that displays the string “Hello, World!”.

Requirements:

- (1) Use the editor program in the computer system to input the source program.
- (2) Compile and link the source program. If errors are found, correct them using the editor and repeat compilation/linking until no errors occur.
- (3) Run the program.

7. Given the three sides a, b, c of a triangle, the area formula is:

$$s = \sqrt{p(p-a)(p-b)(p-c)},$$

where $p = (a + b + c)/2$. Following Example 1.2, write a C program that calculates and outputs the triangle area based on the three sides input from the keyboard.

Requirements:

- (1) Add comments to at least 5 statements in the program (use English or pinyin if Chinese is not supported).
- (2) Use the editor program to input the source program.
- (3) Compile and link the source program, correcting errors until no compilation/linking errors occur.
- (4) Run the program with inputs $a = 3.0$, $b = 4.0$, $c = 7.0$.